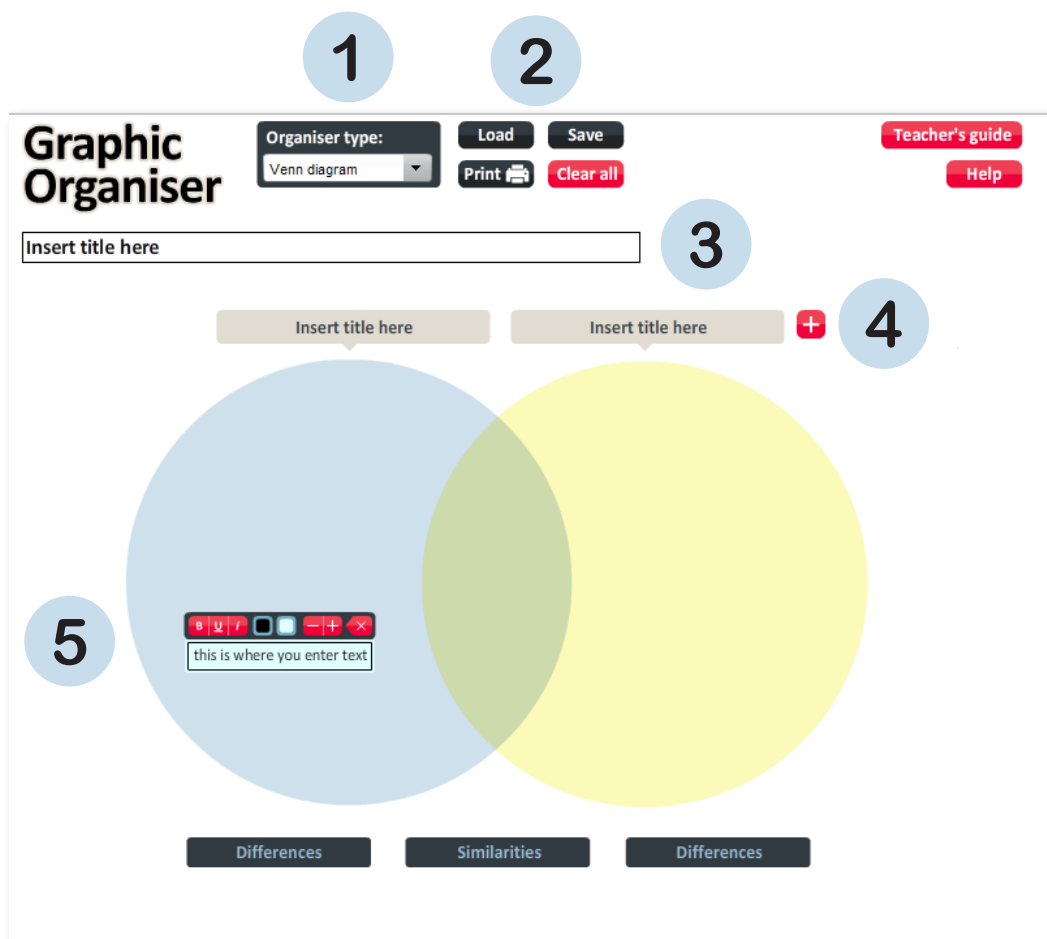


Select a numbered area to find out about the features of the *Graphic organiser toolkit*.



## Selecting graphic organisers

You can use different graphic organisers to suit your needs. It's worthwhile to practise using the different types of graphic organisers for different activities.

| Learning focus                                 | Graphic organiser                     |
|------------------------------------------------|---------------------------------------|
| Brainstorm                                     | Concept map, Y chart                  |
| Compare and contrast                           | Venn diagram, T chart                 |
| Link what you know with what you want to learn | TWLH chart                            |
| Sequence ideas, steps or process               | Flow chart                            |
| Explore relationships                          | Cause and effect wheel                |
| Analyse parts or sections                      | PMI chart, SWOT analysis              |
| Make decisions                                 | SWOT analysis                         |
| Reflect on learning                            | Mind map, Plus/Minus/Interesting tool |



Select the down arrow to view the list of organisers.

Select the organiser type you wish to use.

There are currently 11 graphic organisers to choose from:

- Variable grid
- PROE chart
- TWLH chart
- PMI chart
- SWOT analysis
- T chart
- Y chart
- Venn diagram – 2 circles
- Venn diagram – 3 circles
- Flow chart
- Cycle chart.

### Variable grid

Use this grid to help generate questions for fair testing in scientific investigations. For example: *What happens to the distance the ball rolls when we change the type of surface?*

### Fair tests

A fair test changes just one variable and keeps the others the same. This concept is known as CMS (Change, Measure, Same). When applying a fair test, you follow three rules:

- **Change** one variable (e.g. types of surfaces). This is the independent variable.
- **Measure** one variable (e.g. the distance an object rolls). This is the dependent variable.
- Keep all other variables the **same** (e.g. size of object, colour of object, type of object, shape of object, amount of force used, temperature of the surface, type of surface).

It is important to change only one thing, measure only one thing and keep everything else the same. You could use a mnemonic to remember this concept, such as 'Cows Moo Softly'.

**Note:** A 'control' is also used, in which all the variables stay the same. This way you can make sure that the results accurately show what has caused the change.

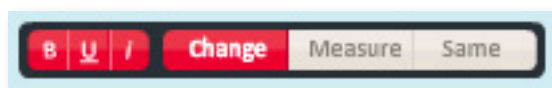
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## How to use the variable grid

First, decide on an investigation and enter it in the 'Insert title here' box.

Next, fill in each text box with a variable. Enter the variable you will measure in the green 'M' box. Enter the variable you will change in the blue 'C' box. Enter all variables that you will keep the same in the grey 'S' boxes. (You do not need to fill in all the boxes.)

Once you have added all your variables to the grid, you can check that they are appropriate and modify which variables you will measure or change. Select the text box that you wish to modify, then use the 'Change, Measure, Same' toolbar that appears to change the properties.



For example, if you select a grey 'Same' box and then select 'Change' from the properties box, the variable will become blue. (Only one variable can be blue and one green.)

**Note:** The green 'Measure' box always moves to the centre of the grid. This is the conventional layout of a variable grid.

As you enter variables into the text boxes in the grid, the fair test question underneath the grid automatically updates. You may need to revise the wording of your variables in the text boxes so that the question makes grammatical sense.

**Note:** A OneChannel on demand demonstration about the variable grid is available on the Learning Place. Use Program key P9002691 to access the demonstration.

## PROE chart (Predict, Reason, Observe and Explain)

Use this chart to record predictions, reasons to support predictions, observations for each prediction and explanations for why the results did or did not match the predictions (based on a comparison of predictions and observations).

## TWLH chart (Think we know, Want to learn, what we Learned, How we know)

Use this chart to elicit students' prior knowledge, determine the questions that students want to know the answer to, document what has been learned, and allow students to include evidence from investigations and learnings to explain and justify claims. It should be filled out before, during and after the unit of study.

## PMI chart (Plus, Minus, Interesting)

Use this chart to consider the benefits, deficits and interesting points of an idea, topic, subject or proposal. It can be used for initial brainstorming, or at the conclusion of an investigation or task.

## SWOT analysis (Strengths, Weaknesses, Opportunities, Threats)

A SWOT analysis is used to analyse and evaluate the strengths, weaknesses, opportunities and threats of an idea. It is useful when you make decisions about implementing actions or new ideas.

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## T chart

Use this chart to organise and/or compare information or thinking about two facets of one topic (e.g. facts and opinions, pros and cons) or about two different concepts or subjects.

## Y chart

Use this chart to describe three aspects of a topic, such as observations (e.g. looks like, sounds like, feels like) or characteristics (e.g. animal's habitat, diet, physical appearance).

## Venn diagrams

Use these diagrams to compare and contrast two or three topics. Organise information and ideas in the circles to show differences between the topics in the parts of the circles that do not overlap. Depending on the type of information you're investigating, you can use two or three circle Venn diagrams.

## Flow chart

Use this chart to organise ideas, information or processes in a sequence.

## Cycle diagram

Use this diagram to show how items, events or stages are related to one another in an ongoing or repeating cycle.

## Menu options

2

Select this button to reload a previously created graphic organiser. Find the file and open it. The graphic organiser will reload.

Select this button to save the graphic organiser at any point. Give the file an appropriate name (e.g. John Smith\_variable grid), choose where you would like to save it (e.g. USB memory stick, 'My documents') and select Save.

Teachers, select this button to access strategies to support higher order thinking.

Select this button to print a blank or completed graphic organiser.

Select this button to undo all the changes you made to the graphic organiser. This will remove the title, headings, labels and text boxes.

Select this button to access these instructions at any time.



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## Insert title

3

Insert title here

There is an option to give each graphic organiser you create a title. Choose a title that best describes the purpose and content of the organiser (e.g. 'Comparing physical and chemical changes'). Type the title directly into the 'Insert title here' field.

## Adding text boxes

4



Select the **red plus icon** to add a text box to your graphic organiser. This text box will be either free-moving (i.e. you can move it anywhere on the graphic organiser) or set in a fixed position, depending on the type of organiser you have selected. In some graphic organisers you can swap text boxes around – just select the text box you want to move and drag it over the one you want to swap it with.

**Note:** The text boxes have already been added to the variable grid, so no red plus icon appears in this organiser.

The following graphic organisers have free-moving text boxes:

- Y chart
- Venn diagram – 2 circles
- Venn diagram – 3 circles.

**Note:** These free-moving text boxes are the only ones that allow you to change the size of the text and box.

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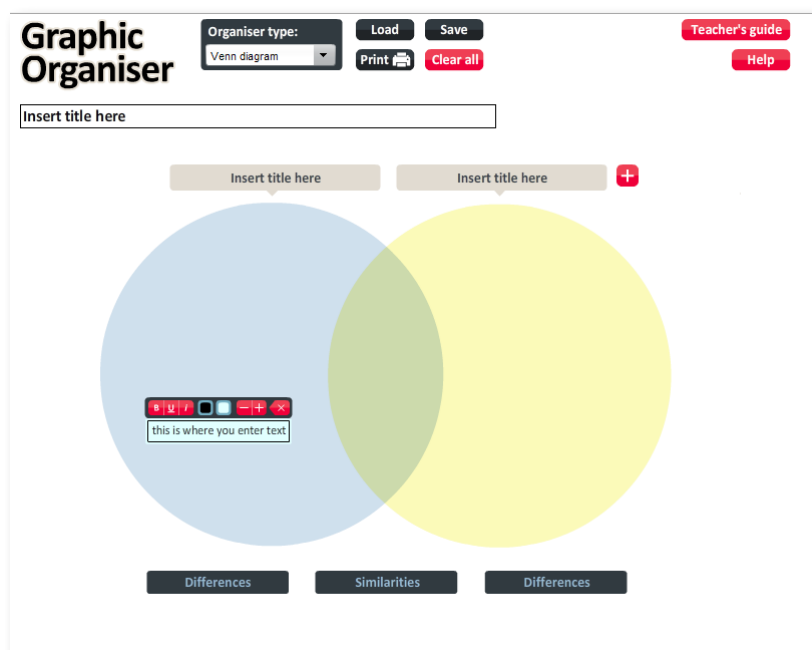
The following graphic organisers have fixed text boxes:

- Variable grid
- PROE chart
- TWLH chart
- PMI chart
- SWOT analysis
- T chart
- Flow chart
- Cycle chart.

**Note:** You can swap flow chart text boxes with each other by selecting and dragging them. You can also change the colour of each box.

## Properties

5



The properties field is used to format text in the graphic organisers. It will become active only after you have added a text box to a graphic organiser and selected it. A black line will appear around the outside of the text box to show that it is active, and you can then start entering text.

Select **B** to **bold** the text.

Select U to underline the text.

Select *i* to apply *italics* to the text.



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Change the text colour by selecting the **black icon** and selecting a colour from the colour palette. Note that this icon will default to the colour you last selected.

Change the colour of the fill in the text box by selecting the **white icon** and selecting a colour from the colour palette. Note that this icon will default to the colour you last selected.

Select the **minus** icon to make the text box smaller. (Text box size changes only apply to free-moving text boxes in Y charts and Venn diagrams.)

Select the **plus** icon to make the text box bigger. (Text box size changes only apply to free-moving text boxes in Y charts and Venn diagrams.)

Select the **cross icon** to delete the current text box.

**Note:** If you wish to delete all changes to the graphic organiser, including all text boxes, you can select **Clear all** from the Menu options.

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